Supplementary Material

Whether Chemical Warfare Agent Simulant Designations Look Supported Depends on Which Compounds Are in the Comparison

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This Supplementary Material carries the acute toxicity axis, the predictive interpolation of that axis, the decoupling test between fidelity and toxicity, the phosphorus-local reactivity analysis, and the test of whether a descriptor space corresponds to measured colorimetric array response. These results stand on their own data and are referenced from the main text where they bear on it; they are placed here because the argument of the main text turns on the composition of the comparison panel and does not depend on them.

S0.1 Safety axis

Acute inhalation toxicity was taken as the EPA AEGL-3 value at 10 min, the concentration above which life-threatening effects or death are predicted, parsed directly from the official compiled table [1]; the derivation of these values for the agents is described by Watson et al. [2], and the concentration–time exponent on which short-exposure extrapolation of such criteria rests was established by ten Berge et al. [3]. A single fixed exposure duration was used rather than a derived concentration–time surface, and the tier value is used directly as a per-compound property rather than as an exceedance probability. Agent degradation products, which carry their own and in places distinct toxicity [4], are outside the compound set. Values published in ppm were converted using $c\,[\mathrm{mg\,m^{-3}}] = c\,[\mathrm{ppm}] \times M/24.45$. Compounds without a value are recorded as such, with the reason, and are never imputed.

Extraction is the step on which every safety-axis number depends, so we describe it in full. The compiled table carries 270 chemical entries that present an AEGL table of their own, besides a holding list of substances for which the agency states that values are not being developed. Reading a ten-minute AEGL-3 from one of them is not simply a matter of taking the first number on the AEGL-3 line, for four reasons that the table itself imposes. Some entries place a footnote marker in the ten-minute cell and give the value below the table, as *AEGL 3 - 10 min = ** 40,000 ppm*, because the value exceeds half the lower explosive limit. Some give values only for the shorter durations and *NR* for the longer ones, so the row carries three numbers rather than five. Some print the agency's own mg $\mathrm{m^{-3}}$ equivalent beside each value, in brackets for the agents and in parentheses elsewhere. And the text layer of the document occasionally breaks one row across two lines. We therefore parse each AEGL row into exactly five duration cells, where a cell is a number, an *NR*, a *see below*, or a bare footnote marker, and refuse to use any row that does not yield five; footnote lines are separated from table rows by grammar before parsing, and the ten-minute value is read from the text following the footnote's equals sign, so the duration labels that precede it cannot be mistaken for values. Of the 270 entries, 261 yield a ten-minute AEGL-3: 252 from a table cell and nine from a footnote. Two of those 261, magnesium aluminium phosphide and a 60% nonafluoroisobutyl ether preparation, carry no registry number in the table and are set aside here, because without one their structure cannot be resolved unambiguously, which leaves 259 (250 from a table cell and nine from a footnote). They are not the two set aside in Section S0.2, which have registry numbers but no single molecular structure.

Two independent checks were applied, and the distinction between them matters. The conversion constant was checked against the agency's own bracketed mg $\mathrm{m^{-3}}$ figures, which it reproduces to within 0.9–4.0% over eight comparisons (GB, GD, GA and GF at AEGL-2 and AEGL-3). That check validates

the arithmetic and nothing else: every agent value carrying a bracketed conversion is below 1 ppm and contains no thousands separator, so a defect in numeric extraction could not appear in it. The extraction itself was therefore checked over the whole table by a second reader that shares neither the text-extraction engine nor the algorithm: it renders the document with a different tool, locates the character columns of the five duration labels in each header line, and reads the ten-minute value from the corresponding substring of the AEGL-3 line. A positional reader of this kind cannot reproduce a token-ordering fault, which is the point of using one. The two readers agree on all 259 entries, on all 257 of those that resolved to a structure and on all nineteen values in the twenty-eight-compound panel, and the check covers all 25 values at or above 1000, where a thousands separator appears. We report the coverage of this check and not only its pass rate, because a validation sample that excludes the failure mode it is meant to detect establishes nothing.

S0.2 Predictive interpolation of the safety axis

To test whether the missing values could be filled, a random forest was trained on $\log_{10}$(AEGL-3, 10 min) using 37 RDKit descriptors, with five random seeds, five-fold cross-validation and 10 000 bootstrap resamples for interval estimates. Of the 259 entries with a ten-minute AEGL-3, 257 resolved to a structure through PubChem; the two that did not are a mixed-isomer entry and a protein toxin, neither of which has a single molecular structure. Five of the 257 are interhalogen compounds that RDKit cannot parse, and two were removed because their stored structure failed the identity check described below, leaving 250. Twenty of these are military chemical agents, identified by registry number rather than by matching on name, and all twenty were held out of training and used only for evaluation, leaving 230 for training. The held-out set is the G series including cyclosarin, VX, sulfur mustard, the three nitrogen mustards, lewisite 1, methyldichloroarsine, ethyldichloroarsine, phenyldichloroarsine, diphenylchloroarsine, adamsite, BZ, phosgene oxime, hydrogen cyanide, phosgene and chloropicrin. Matching on names rather than registry numbers is what makes such a set hard to assemble: the table lists cyclosarin as “GF Agent”, which a pattern written for “Agent G” does not find, and it lists three arsenical vesicants and BZ in rows whose longer durations are *NR*, which a parser that requires five numbers per row discards entirely. External validation on held-out compounds rather than cross-validated statistics alone is the minimum standard for a predictive structure–activity model [5, 6]. An applicability domain was defined as the mean distance to the five nearest training compounds in standardised descriptor space, with the boundary at the 95th percentile of the training set’s own such distances. This is the k-nearest-neighbour distance criterion whose relation to prediction accuracy was established by Sheridan et al. [7] and which appears among the standard formulations surveyed by Netzeva et al. [8] and Roy et al. [9]; we have used the same construction to bound predictions in hazard-prioritisation screens [10]. Whether such a domain can detect a compound that is close in descriptor space but outside the training set’s mechanistic coverage is precisely what Section S0.13 tests, and is related to the open-set problem of recognising an input as unfamiliar rather than assigning it to a known class [11].

S0.3 Decoupling test

Candidates were the AEGL compounds for which structure, electronic descriptors and a measured AEGL-3 were all available, excluding the authentic agents themselves ($n = 207$). Eight further records were set aside under four rules declared in advance, all of which ask the same question: are these electronic descriptors defined for that species at all? A record is excluded if its stored structure has more than one component, because the relative placement of the fragments is arbitrary and the dipole moment is therefore undefined (three records); if the species is open-shell, because the orbital parse reads only one spin manifold, so that its HOMO and LUMO do not carry the closed-shell meaning (three); if it is a bare atom with no bonds, because molecular descriptors carry no meaning (three); or if the HOMO–LUMO gap is below 0.5 eV, because a closed-shell single-reference description does not hold at that gap and the orbital energies themselves stop meaning anything (one). Some records fall under more than one rule. The gap rule bites on a single record: sodium cyanide returns a gap of 0.097 eV where the next smallest gap in the

pool is 1.31 eV, so the 0.5 eV bar separates that one record and no other. Retaining all eight records raises the candidate count to 215 and moves the Spearman coefficient from $+0.062$ to $+0.030$, leaving every conclusion in Section S0.14 unchanged. Fidelity was defined as the standardised Euclidean distance in the three-coordinate electronic space to the nearest authentic agent; safety as the measured $\log_{10}$(AEGL-3). For the quadrant analysis, the safety bar was set at the measured AEGL-3 of dichloromethane, the incumbent simulant, read from the parsed table rather than entered by hand; a replacement simulant should not be more acutely toxic than the compound it replaces. Because that bar turns out to be demanding, the count is also reported at one and two orders of magnitude below it. Association was tested with Spearman and Pearson correlation, with 10 000-resample bootstrap confidence intervals [12]. Effect sizes are reported with intervals rather than p-values alone [13]. Where a paired comparison was made, normality of the paired differences was assessed by the Shapiro–Wilk test [14] and the Wilcoxon signed-rank test [15] used in place of a paired t-test where normality was rejected; those paired tests are reported for completeness only, because, as Section 2.4 explains, they are confounded by group size and the exact permutation test supersedes them. Multiplicity is corrected within each panel configuration. The six distance statistics applied to the three descriptor spaces form one family of eighteen tests on the same descriptor table, and we report the whole family rather than a selection from it, so the Benjamini-Hochberg procedure [16] is applied across those eighteen and the smallest adjusted value is given in Table 1. An earlier version of this work reported one nominal value per space and declined to correct, on the ground that the spaces were separate analyses. That ground fails once the statistic is also chosen by the analyst, and we treat the earlier position as an error rather than a convention.

Defining that family is itself a choice, and a paper arguing that such choices decide the answer cannot leave its own unexamined. We therefore report four schemes rather than one. The eighteen tests are corrected together; the six statistics are corrected within each space separately, treating the spaces as three analyses; a single statistic, the mean difference, is treated as though it had been specified in advance, leaving three tests; and, because the three spaces are nested and the six statistics are functions of one distance matrix, the dependence is handled exactly rather than approximated. All eighteen tests are computed on the same enumeration of designations, so the joint null distribution is available: for each designation we convert every test statistic to its one-sided p within its own null, take the smallest of the eighteen, and read the observed smallest p against that distribution. This is the single-step min-P procedure of Westfall and Young [17], and because the permutations are shared it requires no assumption about the correlation structure. We also report the approximation it replaces, the effective number of independent tests estimated from the eigenvalues of the correlation matrix of the eighteen nulls by the method of Li and Ji [18] used as a Bonferroni divisor, so that the two can be compared. The five answers are given in Table 2 and they do not agree.

S0.4 Dimensionality and subspace controls

The structural and electronic spaces differ in dimension by more than an order of magnitude, and the concentration of Euclidean distances depends on dimension, so a difference between the two spaces is not by itself a difference between chemistries. We bounded the effect in two ways, both reported in Section S0.6. The standardised structural space was truncated to its first three principal components and the permutation test rerun on that three-dimensional space. Separately, 2000 random three-descriptor subsets of the structural space were drawn without replacement (seed `0xC0DE`) and the full exact permutation test run on each, giving the distribution of p that an arbitrary three-dimensional slice of these descriptors produces. The observed structural value is interpretable only against that distribution.

The random-subspace control of Section S0.4 draws three descriptors from a correlated pool, so three descriptors do not carry three degrees of freedom and the resulting rate of nominal significance may be inflated. We repeated it on ZCA-whitened coordinates, obtained by standardising the structural block, taking the eigendecomposition of its covariance, and applying $V\Lambda^{-1/2}V^{\top}$ over the non-null eigenvalues, so that the coordinates drawn from are uncorrelated by construction. Both the raw and the whitened rates are reported. That draw count carries a Monte Carlo standard error near one percentage point on a proportion and resolves a percentile to 0.05.

S0.5 Measured colorimetric array response

Measured response was taken from a 29-dye colorimetric array reported in earlier work from this laboratory [19, 20, 21]. Colour difference ΔE is given per dye for three compound sets at nominal vapour concentrations of 10, 50, 100 and 500 (arbitrary units as reported), in two imaging channels: ultraviolet-excited emission and reflectance. The sets are DMMP with GA, GB, GD, GF and VX; AC, CG, CK, HD, HN and L with PS; and four A-series agents. The first two sets supply the twelve compounds that are also in the descriptor panel: six of the six in the first set and six of the seven in the second, the exception being nitrogen mustard HN, which the descriptor panel does not carry. The third set is excluded entirely and the criterion is arithmetic rather than editorial: none of its four compounds appears in the descriptor panel, so no pair of a measured pattern and a descriptor distance can be formed from it. It contributes to no analysis in this paper.

The three sets are different compounds measured on different plates, not replicates, so no ΔE value is compared across sets; every analysis is run within a set and the results are reported separately or combined only at the level of correlation coefficients. The two channels are likewise not replicates but physically different measurements, and the same compound's patterns in the two channels agree only to a mean cosine of 0.584. They are therefore treated as independent modalities, and agreement between them is used as evidence of reproducibility rather than averaged away.

Each compound's 29-dimensional response at each concentration was scaled to unit length, so that pattern rather than magnitude is compared. The compound-axis mean was then subtracted at each concentration to remove the common non-negative response component, and the resulting vectors were averaged over concentrations and renormalised to give one pattern per compound per channel. Similarity is the cosine between patterns and dissimilarity is one minus that. Confidence intervals are bootstrapped over concentrations with 10 000 resamples, which means they describe variation across the four concentrations of one plate set and not agreement between independent preparations.

Two properties of this processing constrain what the patterns can be used to show, and we state both because the first version of this analysis did not.

Subtracting the compound-axis mean at each concentration makes the resulting vectors sum to zero across compounds, which fixes the mean cosine between distinct compounds at $-1/(n-1)$ for n compounds in a set. A negative between-compound cosine is therefore an identity of the centring and not evidence that the array separates the compounds. We report the observed mean against that floor in Section S0.11 rather than against zero.

Pairwise distances are not independent observations. A set of n compounds gives $\binom{n}{2}$ pairs that share compounds, so correlating measured dissimilarity against descriptor distance over pairs and testing it as if the pairs were independent inflates significance; the unit of analysis is the compound. We therefore test the association by Mantel permutation [22], permuting compound labels on the descriptor distance matrix 10 000 times (seed `0xC0DE`) and comparing the observed Spearman coefficient against $|\rho|$ of the permuted matrices. Both the Mantel value and the value that treating pairs as independent would have given are reported, so the size of the error is visible.

S0.6 Dimensionality and the value of the nominal result

The structural space has 38 coordinates and the electronic space has 3. Euclidean distance behaves differently in the two, so a difference between them could be geometric rather than chemical. We tested that, and it is not. We also tested what the nominal structural value is worth against the descriptors it was drawn from, and there the answer is less comfortable.

Truncating the standardised structural space to its first three principal components does not weaken the result; it strengthens it, from $p = 0.028$ to $p = 0.0044$ on the twenty-eight-compound panel. At matched dimension the two spaces are still 62-fold apart in p (0.0044 against 0.275). Whatever separates the structural from the electronic result, it is not that one space has more coordinates than the other, and we say so because the opposite is the natural suspicion and because an earlier draft of this paper entertained it.

The second control is the informative one. We drew 2000 random three-descriptor subsets of the structural space and ran the full exact permutation test on each. On the twenty-eight-compound panel the median p is 0.094, the 5th to 95th percentile range is 0.006 to 0.710, and 36.3% of the subspaces reach $p < 0.05$ (Monte Carlo standard error 1.08 percentage points). The observed 38-dimensional value of 0.028 sits at the 25.8th percentile of that distribution, and the electronic value of 0.275 at the 77.9th. Both are ordinary draws. The three-principal-component value of 0.0044 sits at the 2.1st percentile, which is what the leading components of a space should do and is not independent evidence.

That rate is partly an artifact of the descriptors themselves, and we report the correction because it is larger than we expected. Three descriptors drawn from a correlated pool do not carry three degrees of freedom. Repeating the control on ZCA-whitened coordinates (Section S0.4), where the coordinates are uncorrelated by construction, the rate falls from 36.3% to 16.9% and the median p rises from 0.094 to 0.295. The correlation structure of the structural block therefore inflates the apparent ease of reaching nominal significance by more than a factor of two. The comparison that the argument rests on is the percentile rather than the rate, and that is unaffected: the observed value is an ordinary draw from the subspaces that were available to be chosen.

So the structural and electronic spaces do differ, and the difference is chemical rather than geometric; what the data do not support is that the difference says anything about the compounds that were designated. A value at the 28th percentile of the subspaces one could have chosen, in a family of eighteen tests whose smallest adjusted value is 0.159, on the smaller of two defensible panels, is not an argument. Picking a space, a panel and a statistic fixes the answer, and the practice this paper examines has never written down which to pick.

S0.7 Weighting the two descriptor blocks equally

The merged space puts 38 structural coordinates and 3 electronic ones into one Euclidean distance. Standardising each descriptor equalises the variance per coordinate, not per block, so the expected contribution is 38 against 3 and the merged distance is in practice a structural distance. The main text says so; this section quantifies it, because a reader is entitled to ask whether the merged result is an artifact of that imbalance. A colleague who read a draft asked exactly this.

We repeated the whole analysis on a distance that equalises the expected contribution of the two blocks,

$$d^2_{\mathrm{bal}} = \frac{1}{n_s}\sum_{i=1}^{n_s}(\Delta z_{s,i})^2 + \frac{1}{n_e}\sum_{j=1}^{n_e}(\Delta z_{e,j})^2, \tag{1}$$

with $n_s = 38$ and $n_e = 3$, leaving the structural and electronic spaces untouched since each is a single block. The same exact enumeration was used, over the same three configurations, with the same six statistics.

Table S1: Equal weight per descriptor against equal weight per block, in the merged space. Valid-match counts are identical under the two weightings. The smallest p is taken over the eighteen-test family of three spaces by six statistics, and q is the Benjamini–Hochberg adjusted value of that family.

	Valid	Smallest nominal p		Smallest q	
Configuration	matches	per descriptor	per block	per descriptor	per block
Thirty-five, incumbent three	0/11	0.151	0.076	0.617	0.617
Thirty-five, plus literature eight	5/11	0.039	0.021	0.212	0.222
Twenty-eight, incumbent three	3/11	0.025	0.009	0.159	0.132

Two things follow, and they point in opposite directions. The valid-match counts do not move at all: zero of eleven on the primary panel, five with the literature eight admitted, three on the twenty-eight-compound panel, exactly as reported in the main text. The paper's headline is therefore invariant to the weighting, and so is the pre-specified-statistic counterfactual of Table 2, which stays at 0.042 on the twenty-eight-compound panel: the merged nominal value there falls from 0.025 to 0.015, but the value

that binds after correction is the structural one, which the reweighting does not touch. No configuration has a Benjamini–Hochberg adjusted value below 0.05 under either weighting.

Every nominal p falls, however, and that is why the main text states the primary-panel result as a conclusion about significance rather than as a numerical floor. The smallest value of the eighteen-test family there is 0.151 with equal weight per descriptor and 0.076 with equal weight per block, so a floor quoted from either weighting would not hold under the other. The direction is away from the designations: equalising the blocks makes the incumbent simulants look better on the nominal scale while leaving them with no agent matched. Block weighting is therefore a fourth choice of the same kind as the panel, the space and the statistic, and it was not declared in advance either. We report it because we did not pre-register it, and because a reader who did would have obtained numbers we would not otherwise have published.

S0.8 A chemically grounded lewisite simulant

Of the eleven agents, lewisite is the one whose incumbent simulant is chemically least like it: dichloromethane shares neither the central element nor the halogen count. A reader may reasonably ask whether the lewisite result reports a failure of the designation or only a poor choice of comparator on our part. We therefore repeated the panel analysis with arsenic trichloride added. The compound is used as a lewisite simulant in the sensing literature [23]. We found no designating document for it, so it enters as a literature simulant, on the same footing as the other five, and not as an incumbent. Descriptors were generated by the same procedure as the rest of the panel, without modification. The optimised geometry is pyramidal, with Cl–As–Cl angles of 98.8° against a reported 98.4°.

Table S2: Valid matches and the lewisite row across panel configurations. A match requires a normalised distance of at most two *and* that no non-simulant lies closer.

Configuration	Valid matches	Nearest simulant to lewisite	Normalised distance
Twenty-eight, incumbent three	3/11	dichloromethane	2.13
Thirty-five, incumbent three	0/11	dichloromethane	2.38
Thirty-five, plus literature eight	5/11	CEES	2.35
Thirty-six, incumbent three	0/11	dichloromethane	2.04
Thirty-six, plus literature	6/11	$AsCl_3$	1.25

The incumbent-three count is unchanged at zero of eleven. This is not a coincidence of these data. A compound that is not a designated simulant can enter the matching rule only through its second clause, as one more non-simulant that a designated simulant must beat; adding one can make a match harder and never easier, and the count is already at its floor.

The single match the addition creates is lewisite's. Under the literature set without arsenic trichloride the nearest literature simulant to lewisite is 2-chloroethyl ethyl sulfide, a sulfur mustard simulant, at 2.35 and outside the threshold: the five contained no candidate built on the right element. Arsenic trichloride enters at 1.25 and matches. We draw no recommendation from this, for the reason given in the main text. The point is narrower. For one agent the compound the descriptor space selects and the compound that carries the designation are different compounds, and the designation is the one for which we found no document.

Limitation. The semiempirical method returns a dipole moment of 0.06 D for arsenic trichloride against a reported value near 1.6 D. The method is applied identically to every compound in the panel, so relative positions remain comparable, but absolute electronic descriptors for heavy main-group compounds should not be read on their own.

S0.9 What the test could have detected

A null result invites the question whether the design could have seen an effect at all, and the exact enumeration answers it without further assumption. We moved the incumbent simulants closer to the agents by δ, measured in typical nearest-neighbour separations of the same space, left the rest of the panel untouched, and read the shifted statistic against the null built from the unmodified distances. The reading is conditional: on this panel, with only the designated three made more faithful, the test returns $p < 0.05$ at $\delta = 0.22$ (Figure S1). At $\delta = 0$ it returns 0.2831 on the primary panel and 0.0250 on the twenty-eight-compound panel, reproducing Table 1.

The comparison that matters is between that value and the deficit the incumbent compounds actually carry. Across the eleven agents the normalised difference between the nearest incumbent simulant and the nearest other compound has a median of 0.463, more than twice the detectable value. Only at sarin, where it is 0.117, does it fall below what the test can resolve, and that is the agent at which the descriptor bootstrap of Section 3.1 is also weakest, recovering DMMP as nearest in 37.8% of replicates. The null on the primary panel is therefore a statement about the data rather than about the sensitivity of the design, with sarin the one case where the two cannot be separated.

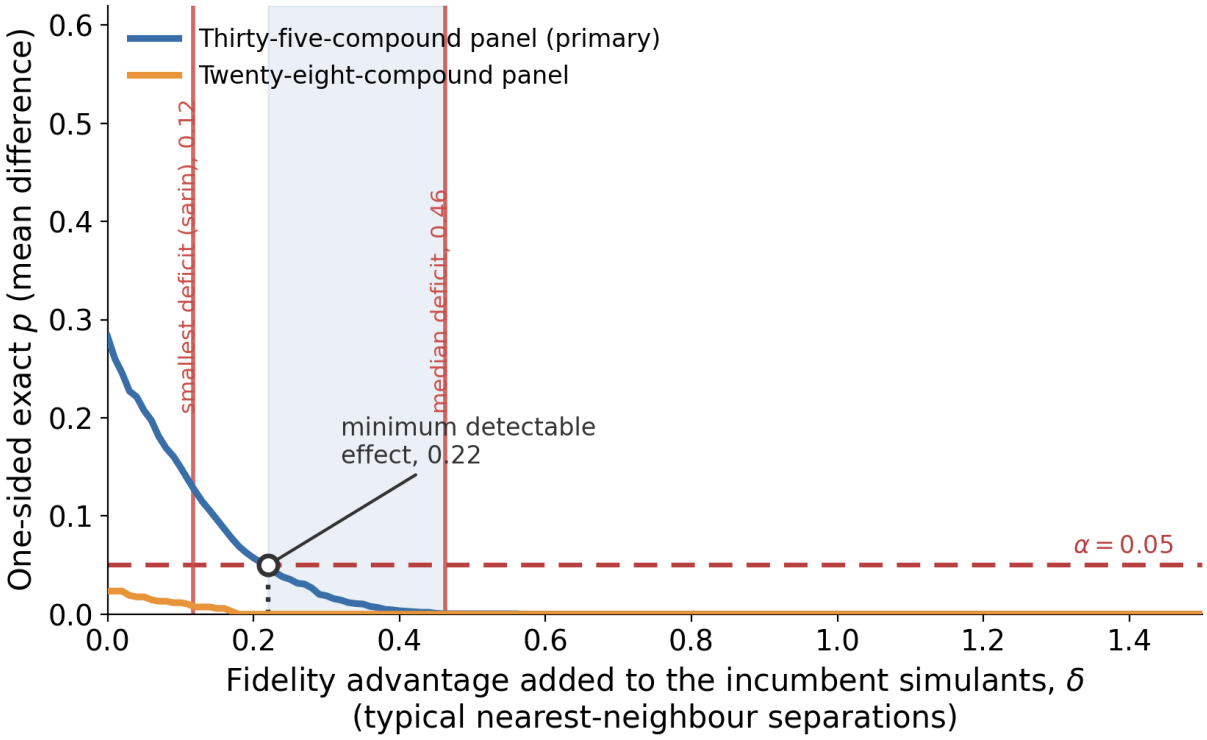


Figure S1: What the permutation test could have detected. One-sided exact p for the mean-difference statistic as the incumbent simulants are moved δ typical nearest-neighbour separations closer to the eleven agents, with the rest of each panel unchanged and the null taken from the unmodified distances. On the primary panel the test crosses $\alpha = 0.05$ at $\delta = 0.22$. The shaded band spans that value and the median deficit the incumbent compounds actually carry, 0.46; the deficit falls below the detectable value only at sarin.

S0.10 Local reactivity at phosphorus

Table S3 gives the phosphorus-centred quantities. The relevant reactivity is the attack of the serine hydroxyl of acetylcholinesterase on phosphorus, so the quantity of interest is the susceptibility of that centre to a nucleophile, which is what $f^{+}(\mathrm{P})$ measures.

$f^{+}(\mathrm{P})$ orders GB > GD > GF > DMMP > GA > VX. DMMP reaches 84% of sarin's value but lies

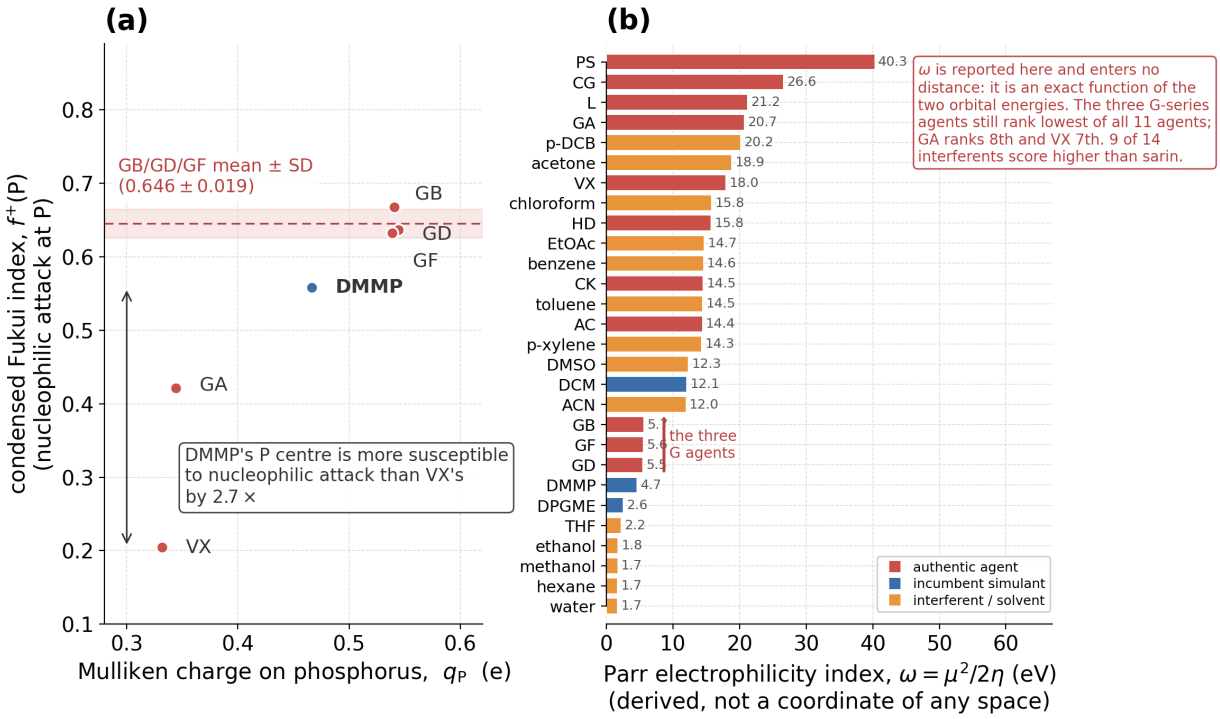


Figure S2: Two quantities that sit outside every distance used in this work. (a) Local electrophilicity against Mulliken charge at phosphorus for the six phosphorus-bearing compounds (Section S0.10). (b) The Parr electrophilicity index, an exact function of the orbital energies that enter the electronic space, which itself enters no distance anywhere in this work (Section 2.3) and on which the three nerve agents rank lowest of the eleven agents. The principal components of the two spaces are in Figure 2a and b.

4.47 standard deviations below the tight GB/GD/GF cluster (0.6456 ± 0.0195), so it is the closest available compound rather than a member of the group. Against VX, DMMP's $f^+(\mathrm{P})$ is 2.73 times larger.

That last ratio depends on which quantity is taken to represent local reactivity, and we report both because they disagree. The composite local electrophilicity of Domingo et al. [24], $\omega_k = \omega f_k^+$, weights the condensed Fukui index by the global index, and under it DMMP reaches 0.69 of sarin and 0.71 of VX rather than 2.73 times VX: the direction of the DMMP–VX comparison reverses. The reversal is carried entirely by the global factor, which for VX is 17.99 eV because the sulfur lone pair raises the HOMO, and for tabun 20.73 eV because of the nitrile. Those are the same values that the next paragraph shows to be unusable as a reactivity ordering for this compound class, so we do not treat ω_k as the arbiter; but a reader applying the standard composite to Table S3 would reach the opposite conclusion about VX, and that is worth stating plainly. Against the GB/GD/GF cluster the two conventions agree and ω_k is the harsher of the two, placing DMMP 6.87 standard deviations below the cluster against 4.47 for $f^+(\mathrm{P})$ alone.

The measured array response bears on this and narrows it. Section S0.11 compares DMMP's 29-dye response pattern with those of the authentic agents. DMMP is closer to sarin than to VX by 1.35 in cosine in the ultraviolet channel and by 0.84 in reflectance, both intervals excluding zero, so on the VX comparison the measurement sides with $f^+(\mathrm{P})$ against ω_k. It does not support the wider reading that DMMP stands in for the G series as a class: GD and GF are both further from DMMP than VX is.

The Wiberg P=O bond order spans 1.9246–2.0146 across all six compounds and does not separate the G series from DMMP; it carries no useful discrimination here. The global electrophilicity index is actively misleading: GD (5.54), GF (5.62) and GB (5.69) are the three lowest values among the eleven authentic agents, and nine of the fourteen interferents score higher than sarin. The reason is that ω responds to the lowest π^* orbital wherever it sits, the nitrile in tabun or the sulfur lone pair raising the HOMO in VX, and it has no particular connection to phosphorus. Global electrophilicity should not be used as a reactivity axis for this compound class.

Table S3: GFN2-xTB local reactivity at phosphorus for the six phosphorus-bearing compounds. $f^+(\mathrm{P})$ is the condensed Fukui index for nucleophilic attack at phosphorus. ω is the global electrophilicity index in the definition of Parr et al. [25] with η equal to half the HOMO–LUMO gap, and $\omega_k = \omega f^+(\mathrm{P})$ is the composite local electrophilicity of Domingo et al. [24]. The two reactivity conventions disagree about VX and the text says why.

Compound	q_{P} (e)	$q_{\mathrm{P=O}}$ (e)	$W_{\mathrm{P=O}}$	$f^+(\mathrm{P})$	ω (eV)	ω_k (eV)
GB	0.54044	-0.42847	1.9527	0.66793	5.69	3.80
GD	0.54460	-0.43006	1.9507	0.63692	5.54	3.53
GF	0.53868	-0.42662	1.9546	0.63203	5.62	3.55
DMMP	0.46630	-0.44588	1.9246	0.55852	4.66	2.60
GA	0.34433	-0.38982	2.0146	0.42143	20.73	8.74
VX	0.33196	-0.40303	1.9863	0.20431	17.99	3.68

GB/GD/GF mean $f^+(\mathrm{P})$ = 0.6456 $\pm$ 0.0195, DMMP 4.47 SD below; mean ω_k = 3.63 $\pm$ 0.15, DMMP 6.87 SD below. ω and ω_k are reported here and used in no distance (Section 2.3). DMMP/VX = 2.73 by $f^+(\mathrm{P})$ but 0.71 by ω_k. Values are GFN2-derived and relative within one protocol.

S0.11 What the array actually measures

Every argument above rests on descriptor distances, and the premise that one of those spaces corresponds to what a vapour-phase sensor reports has so far been assumed rather than shown. We test it against measured response from a 29-dye colorimetric array, reported for three compound sets at four concentrations in two imaging channels, ultraviolet-excited and reflectance (Section S0.5). Twelve of the measured compounds are in the panel: DMMP and the five nerve agents in the first set, and six of the remaining agents in the second.

The array discriminates, but only after the common-mode response is removed, and less of that separation is empirical than the raw numbers suggest. Colour difference is a non-negative quantity, so every analyte shares a common "something responded" component and cosine similarity between raw profiles is compressed towards unity. Between-compound similarity of 0.77 to 0.80 is not separable from the within-compound value of 0.79 to 0.83 across concentrations ($p = 0.06$ to 0.94). Subtracting the compound-axis mean at each concentration separates them, to $+0.35$ to $+0.54$ within compounds against -0.10 to -0.33 between them, in all six combinations of compound set and channel ($p = 8 \times 10^{-5}$ to 0.025).

The between-compound half of that contrast is largely arithmetic. Centring forces the mean cosine between distinct compounds to $-1/(n-1)$, which is -0.200 for the six-compound sets and -0.167 for the seven-compound set (Section S0.5). The observed means clear that floor by 0.013 to 0.066: -0.187 and -0.167 for the first set in the two channels against a floor of -0.200, and -0.101 and -0.117 for the second against -0.167. The negative between-compound similarity is therefore not evidence of separation. The within-compound reproducibility across concentrations, which carries no such constraint, is the part of this comparison that remains informative. All results below use the centred patterns, with that caveat attached.

The measurement bears on Supplementary Section S0.10, and it narrows the claim rather than confirming it. What replicates across both channels is specific to sarin. DMMP's pattern is closer to GB than to VX by 1.347 in cosine in the ultraviolet channel (95% CI 1.062 to 1.436, bootstrapped over concentrations) and by 0.841 in reflectance (0.462 to 0.903), and closer to GB than to the mean of GD, GF and VX by 1.284 and 0.984 in the two channels, every interval excluding zero. The class-level statement is weaker. Taking the mean of the three matched G agents against VX gives 0.513 (0.131 to 0.722) in the ultraviolet channel but 0.137 (-0.144 to 0.290) in reflectance, and the least similar agent to DMMP is GF rather than VX in both channels (-0.721 and -0.496 against -0.702 and -0.273 for VX).

The full ranking contains one result that the descriptor analysis does not anticipate, and we give the whole ranking rather than the comparison we set out to make. In the ultraviolet channel the agent nearest to DMMP is not sarin but tabun, at $+0.764$ against sarin's $+0.645$, with soman, VX and cyclosarin following at -0.492, -0.702 and -0.721. In reflectance sarin leads at $+0.569$ and tabun falls to -0.112. Tabun

is not a match to DMMP under the descriptor criterion of Section 3.3: its normalised distance is 2.03, outside the threshold of 2. So the one measured channel in which the ordering is clearest puts at the top an agent the descriptor analysis excludes, and the agreement between the two channels on tabun is the worst of any agent in the set. We report this as a disagreement between the descriptor table and the measurement and not as a correction to either, because two channels on one plate set cannot settle which is right.

The descriptor analysis assigns GB, GD and GF to DMMP. The array places sarin first in one channel and tabun first in the other, and does not place GD or GF closer to DMMP than VX in either. A sensor chemistry calibrated on DMMP is therefore being shown something that behaves like one G agent at a time, not like the G series, and the descriptor match for the other two does not survive contact with the measured response.

It does not answer the larger question in either direction. Pattern dissimilarity, taken as $1 - \cos$ and correlated against descriptor distance, gives the same inconsistent picture as the permutation tests of Section 3.1. Tested by Mantel permutation over compounds (Section S0.5), two of the eight panel-channel-space cells reach a nominal $p < 0.05$, and they favour different spaces: the structural space in the second compound set's ultraviolet channel ($\rho = +0.539$, Mantel $p = 0.012$) and the electronic space in the same set's reflectance channel ($\rho = +0.571$, Mantel $p = 0.047$). The first compound set gives nothing in either space or channel ($|\rho| \leq 0.225$, Mantel $p \geq 0.22$). The eight cells are one family, and we correct them as we correct the permutation tests of Section 3.1: under Benjamini-Hochberg the smallest adjusted value is 0.098 and no cell reaches $q < 0.05$. We state the nominal values as well because two cells pointing at different descriptor spaces is itself the pattern worth seeing, but neither cell is a result.

Treating the fifteen pairs as independent observations, which the first version of this analysis did, moves these values by up to 0.026 and in both directions: it would have reported the structural cell at $p = 0.038$ rather than 0.012 and the electronic cell at $p = 0.026$ rather than 0.047. The error is not large here because the sets are small, but it is the error that would grow with panel size, and it runs toward significance in the cell with the larger coefficient.

Pooling the two compound sets by Fisher transformation, which is legitimate because they are different compounds on different plates, the structural space gives $\rho = +0.317$ (95% CI -0.072 to $+0.622$) in the ultraviolet channel and $+0.168$ (-0.227 to $+0.515$) in reflectance, while the electronic space gives -0.110 (-0.471 to $+0.281$) and $+0.327$ (-0.061 to $+0.629$). Every interval contains zero. The electronic space is the better predictor in one of the four cells.

Six compounds per set bound how much that null can carry. At $\alpha = 0.05$ and 80% power the design detects $|\rho| \geq 0.669$ from fifteen pairs and 0.492 pooled, and the Mantel null is built from the 6! relabellings of six compounds, so the smallest attainable one-sided p is of order 10^{-3} and the resolution is coarse. These data exclude a strong association between either descriptor space and the measured response and nothing weaker. They do not show that the electronic space governs sensor response, do not place it above the structural space, and do not show it to be irrelevant. The claims in the rest of this paper are written to that limit.

S0.12 The measured safety axis, and the holes in it

Nineteen of the twenty-eight compounds have an AEGL-3 at 10 min (Fig. 3a). The ordering runs VX (0.0295 mg m^{-3}), soman (0.365), sarin (0.367), cyclosarin (0.391), tabun (0.729), sulfur mustard (3.84), lewisite 1 (3.9), chloropicrin (13.4), phosgene (14.6), hydrogen cyanide (29.8), acetonitrile (403), chloroform (19 530), benzene (30 990), p-xylene (31 264), toluene (37 685), acetone (38 007), dichloromethane (41 685), n-hexane (42 296) and methanol (52 420). The axis spans six orders of magnitude, and the incumbent simulant dichloromethane sits at the safe end of it, 1.4×10^6 times less acutely toxic than VX and 1400 times less toxic than hydrogen cyanide. On this axis the incumbent simulant is not the weak link.

It is, however, not the safest thing on the panel. Methanol and n-hexane both carry higher AEGL-3 values than dichloromethane, and neither is used as a simulant for anything. The point is not that either should be: it is that the safety axis cannot be what selected the incumbent, because two ordinary laboratory solvents on the same panel score better on it. Against the 207 AEGL compounds that carry

both axes (Section S0.14) dichloromethane sits at the 96th percentile of measured acute inhalation safety, with seven compounds above it. Taking the incumbent as the bar a replacement must meet therefore sets a demanding bar, and the consequences are worked through in Section S0.14.

Nine compounds have no value. Eight are absent from the AEGL list altogether: DMMP, tetrahydrofuran, ethyl acetate, water, DPGME, ethanol, DMSO and p-dichlorobenzene. The ninth, cyanogen chloride, is the one military agent among them, and its absence is a recorded decision rather than an oversight: it appears in the compiled table's holding list, where the agency states that values are not being developed. Checked by CAS against the source document, it is the only one of the nine on that list. DMMP's position is worth stating precisely, because it is the compound that carries all three valid matches found above. EPA's provisional peer-reviewed assessment declined to derive an inhalation reference concentration, recording that the data are insufficient, and derived no inhalation unit risk; no acute inhalation LC_{50} is cited; IRIS and HEAST contain no RfD, RfC or cancer value. What exists is an oral provisional RfD of 0.06 mg kg^{-1} day^{-1} carrying an uncertainty factor of 3000, and an oral cancer slope factor of 1.7×10^{-3} (mg kg^{-1} day^{-1})$^{-1}$ based on mononuclear cell leukaemia in male rats, with the evidence characterised as suggestive [26]. The inhalation safety of the field's principal nerve-agent simulant has not been established; it has not been assessed. The two incumbent simulants therefore sit at opposite ends of the evidence, not of the hazard: for dichloromethane there is a regulator-derived value and it is favourable, and for DMMP there is a regulator's recorded decision not to derive one.

Cyclosarin carries its own entry in the compiled table under CAS 329-99-7, with a ten-minute AEGL-3 of 0.053 ppm bracketed by the agency as 0.38 mg m^{-3}; our conversion from the published parts-per-million value gives 0.39 mg m^{-3}, placing it between sarin and tabun. The entry is titled *GF Agent* rather than *Agent GF*, which is why matching on compound names rather than on registry numbers can miss it; Section S0.1 describes the CAS-keyed procedure used here.

S0.13 A QSAR cannot fill the gaps

On the training distribution the model works. The random forest (Fig. 3c and d) reached $R^2 = 0.458 \pm 0.014$ and RMSE 0.935 in $\log_{10}$ units (95% CI 0.830–1.020) against a training-mean baseline of 1.271 (1.175–1.365) and a logP-plus-molecular-weight linear baseline of 1.225. The intervals for the model and the mean baseline do not overlap, and the out-of-fold error over the whole training set is unbiased, averaging $+0.001$ log units. Nothing in the cross-validated statistics would warn a user off.

On the twenty agents held out of training it failed, in one direction and without exception. Every one was predicted safer than measured, by between 0.40 and 3.42 log units, a factor of 2.5 to 2600. Sarin was placed at 29 mg m^{-3} against a measured 0.367 and VX at 2.4 against 0.0295, errors of 79-fold and 80-fold. Fourteen of the twenty lie inside the applicability domain, so for those the domain check raises no objection, and the worst of those fourteen is still wrong by 2.68 log units, a factor of 482. DMMP's domain distance, 3.28, sits among the agents' own, which is to say inside the region where the model is wrong by one to two orders of magnitude, so its prediction is not used.

The explanation is not that the mechanism is missing from the training set. We checked, because that was our first hypothesis and it is wrong. Thirteen of the 230 training compounds carry an acetylcholinesterase-inhibiting pharmacophore, identified by substructure rather than by name: ten organophosphorus esters (phorate, phosphamidon, monocrotophos, dicrotophos, fenamiphos, parathion, methyl parathion, methamidophos, dichlorvos, malathion) and three carbamate esters (aldicarb, oxamyl, methomyl). The mechanism by which the nerve agents kill, phosphylation of the serine hydroxyl of acetylcholinesterase and the cholinergic crisis that follows [27, 28, 29], is therefore represented, and one of the thirteen, phorate at 0.22 mg m^{-3}, is more acutely toxic than sarin. The model sees the mechanism and still cannot place the agents.

What it cannot do is reach either extreme. Over the whole training set the out-of-fold error is centred on zero, but in the most toxic decile (the 23 compounds at or below 3.21 mg m^{-3}) it averages $+1.41$ log units and is positive for every single compound, while in the least toxic decile (the 23 at or above 8094 mg m^{-3}) it averages -1.13 log units and is negative for all but one. The same symmetry appears inside the anticholinesterase subset taken on its own: phorate, the most toxic of the thirteen, is predicted

177 times too safe, while malathion, the least toxic, is predicted 269 times too toxic. This is regression toward the mean at both tails, which is what a variance-minimising ensemble does with sparsely populated extremes, and the agents are all in one of them.

That diagnosis matters for the applicability domain rather than for the model. A descriptor-distance domain asks whether a compound is unusual in the space of inputs. The failure here is defined in the space of the response, and the two do not coincide: fourteen of twenty agents are ordinary-looking inputs with extreme outputs. An input-space criterion cannot detect an output-space bias, so it certifies exactly the predictions that are worst. This is the same failure, in the safety axis, that Section 3.1 documents in the fidelity axis. A distance in one space is being used to vouch for a property that lives in another.

S0.14 Fidelity and safety are not monotonically associated

Across the 207 candidates carrying both measured axes (Fig. S3a), fidelity and safety are uncorrelated: Spearman $r = +0.062$ (95% CI -0.077 to $+0.199$), Pearson $r = +0.072$ (95% CI -0.092 to $+0.225$, $p = 0.30$), with fidelity accounting for 0.5% of the variance in toxicity (Table S4). The Spearman interval rules out any monotone association stronger than $|r| = 0.199$ and the Pearson interval any linear association stronger than $|r| = 0.225$, so this is a bounded negative result rather than an underpowered one. Resembling an agent electronically does not require being as dangerous as one.

Table S4: Decoupling test. Fidelity is the standardised electronic-space distance to the nearest authentic agent; safety is the measured EPA AEGL-3 (10 min).

Quantity	Value
Candidates with both axes measured	207
Spearman r	$+0.062$ (95% CI -0.077 to $+0.199$)
Pearson r	$+0.072$ (95% CI -0.092 to $+0.225$), $p = 0.30$
Variance of toxicity explained by fidelity	0.5%
Safety bar: dichloromethane, 41,685 mg m^{-3}	$\log_{10} = 4.62$
Faithful and safe at that bar	0
Faithful and safe one order of magnitude below it	8
clearing the structural-alert gate	8
more faithful than dichloromethane itself	8
Most faithful of them	Trichloroethylene (0.25 vs 0.78)

Setting the safety bar at the incumbent simulant's own measured value empties the faithful-and-safe quadrant outright: of the 207, not one compound combines top-quartile fidelity with an AEGL-3 at or above dichloromethane's 41 685 mg m^{-3}, and dichloromethane does not either, its own fidelity of 0.78 falling outside the 0.71 quartile boundary. That is a statement about how safe dichloromethane is, not about how scarce candidates are. Eight compounds in the whole pool reach or exceed it on the safety axis, the incumbent among them, and not one of the eight is faithful.

Lowering the bar by one order of magnitude, to 4168 mg m^{-3}, changes the answer. Eight compounds then combine top-quartile fidelity with that level of measured acute inhalation safety, every one of them clears the structural-alert gate, and every one of them is more faithful than dichloromethane itself. In order of fidelity they are trichloroethylene (0.25), HCFC 141b (0.34), 1,1,1-trichloroethane (0.54), trifluorochloroethylene (0.58), tetrachloroethylene (0.61), hexafluoropropylene (0.63), methyl ethyl ketone (0.66) and vinyl chloride (0.70), every one of them ahead of dichloromethane's own 0.78. The best of them, trichloroethylene, is 3.1 times closer to the nearest authentic agent in the electronic space while being only 1.3 times more acutely toxic by inhalation. At two orders of magnitude below the incumbent the count rises to fifteen, all of which clear the gate.

These are candidates on the axes measured here, and that qualification is not a formality. Trichloroethylene and tetrachloroethylene carry serious chronic and carcinogenic hazard that an acute inhalation criterion does not express, and the chlorofluorocarbons among them raise questions of atmospheric persistence rather than of laboratory safety. What the screen establishes is narrower and still

useful: in the electronic space, compounds more faithful than the incumbent exist at comparable acute inhalation safety, and they were not found by relaxing the chemistry but by looking in the right descriptor space.

One limitation bounds this part of the result. The candidate pool is the set of chemicals for which AEGL values exist, and chemicals receive AEGLs because they are hazardous. Ordinary safe solvents are largely absent from the pool by construction, which depresses the count at any bar. The seven compounds that exceed dichloromethane on the safety axis illustrate the point from the other side: two are hydrocarbon gases, one a refrigerant, one a monomer, and the three a laboratory might actually reach for sit far out on the unfaithful side of the electronic axis, methanol at 2.00 against the 0.71 boundary, well outside the top quartile that defines a faithful candidate.

S0.15 The structural filter

Applied to the 28-compound set (Fig. S3b), the filter assigns Tier A to nine of the eleven authentic agents and Tier B to phosgene, catching ten of eleven; chloropicrin carries no agent-class motif and is not caught. There are no false alerts among the fourteen interferents. DMMP is Tier C: it carries the phosphoryl ester but not a leaving group, which is precisely why it is used as a simulant and why it is noted rather than excluded. Dichloromethane and DPGME raise no alert.

S1 Further discussion of the toxicity axis

The decoupling result is what makes the rest actionable. Because fidelity and acute toxicity show no monotonic association across the compounds where both are measured, the pursuit of a simulant that is both faithful and safe is not a pursuit along a trade-off curve, and the screen bears that out. Against the incumbent's own safety value the quadrant is empty, which says more about dichloromethane than about the chemistry: it is safer than all but seven of the 207 compounds in the pool, and not itself faithful enough to fill the quadrant it empties. One order of magnitude below that bar there are eight candidates, all of them clearing the structural-alert gate and all of them more faithful than the incumbent, with trichloroethylene 3.1 times closer to the nearest agent in the electronic space at 1.3 times the acute inhalation toxicity.

We are not proposing trichloroethylene as a simulant. It carries chronic and carcinogenic hazard that the axis used here does not express, and a replacement chosen on two measured axes still has to survive the others. The claim is narrower and, we think, more useful: the field's incumbent choice is not on the efficient frontier of the two axes it is usually defended on, and better points on those axes are already present among compounds with measured toxicity. That the pool consists of chemicals assessed because they are hazardous depresses every count reported here, so the seven are a lower bound rather than an inventory. The constructive reading is that designing candidates and then measuring them remains worthwhile, and that the filter excluding agent-class motifs by construction can be applied from the start.

Limitations specific to this material. The safety axis is acute inhalation lethality at ten minutes and does not represent chronic, dermal or ocular hazard, nor flammability, reactivity or environmental persistence, which is the main reason the eight candidates of Section S0.14 are presented as positions on two axes rather than as recommendations. Four phosphorus-local descriptors exist for six compounds only and are reported without being used in any 28-compound analysis. Two features of the regulatory source are carried into our numbers and should be read with them. Five of the nineteen panel values, and nine of the 259 in the wider set, are the agency's footnote values for exposures above half the lower explosive limit; they are the agency's ten-minute AEGL-3 for those compounds, but at concentrations where explosion rather than toxicity is the governing hazard, which is a constraint on using them as working exposure limits even though it does not affect their use here as a relative toxicity scale. And the panel's p-xylene is matched to the compiled table's *Xylenes* entry (CAS 1330-20-7), which covers the mixed isomers rather than the para isomer specifically; p-xylene's descriptors are its own, but its safety-axis value is a class value.

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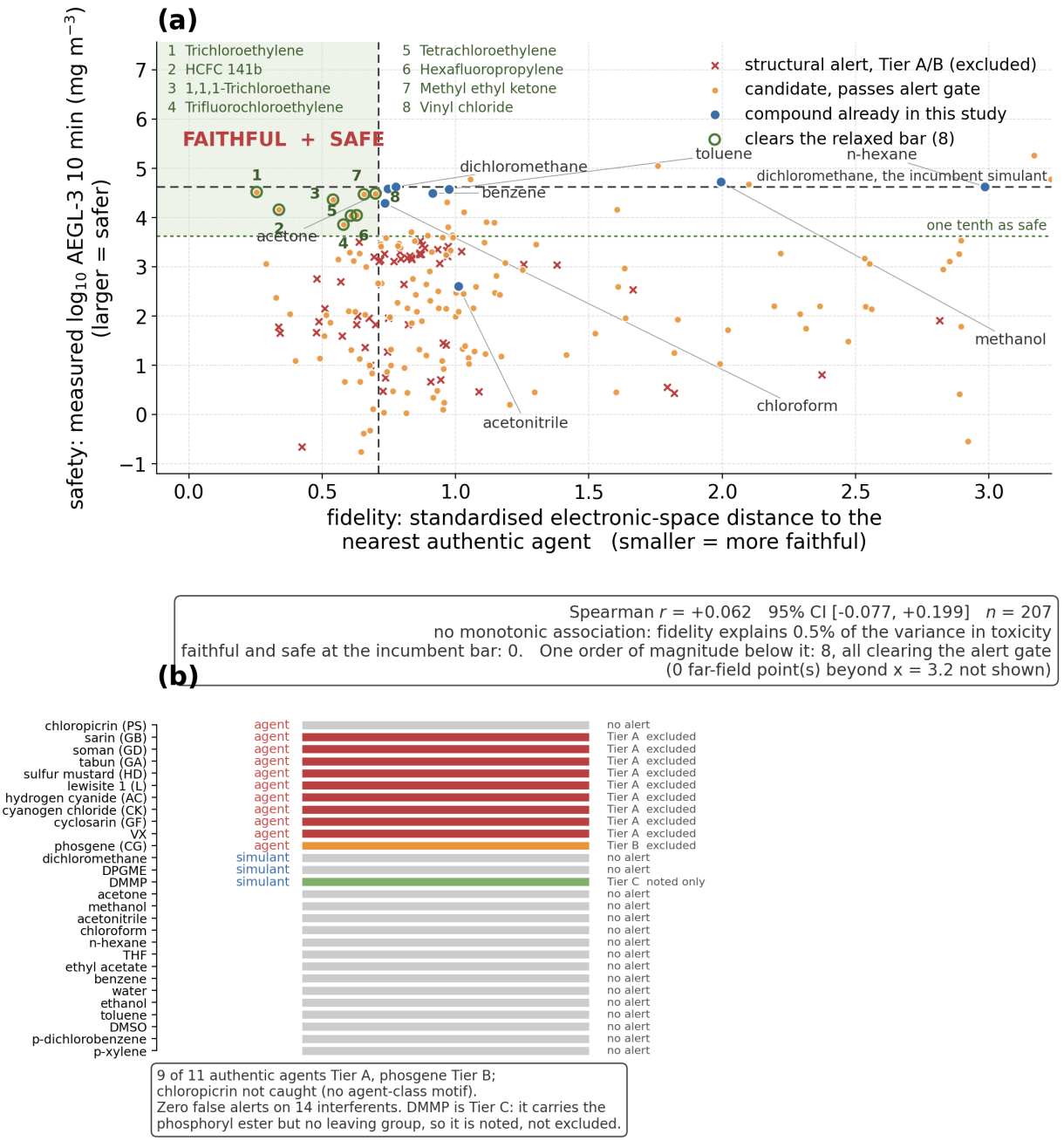


Figure S3: Fidelity and safety are not monotonically associated. (a) Electronic-space distance to the nearest authentic agent against measured AEGL-3 for the 207 candidates carrying both axes; crosses are compounds excluded by the structural filter. The upper dashed line is the measured AEGL-3 of dichloromethane, the incumbent simulant; the dotted line is one order of magnitude below it, and the eight circled compounds are those that clear the lower bar while also falling in the most faithful quartile, numbered by fidelity and named at the top left. The shaded quadrant, faithful and safe at the incumbent's own bar, is empty. (b) The filter applied to the twenty-eight-compound set: nine of eleven authentic agents are Tier A and phosgene Tier B, there are no false alerts among the interferents, and DMMP is Tier C.